

Ablation Workshop Test Case 2.2

1 Description

2 Thermo-chemical Decomposition Model

The pyrolysis model is written in terms of reaction extents $\chi_i(t)$, $i = 1, \dots, N_r$. For each reaction,

$$\frac{\partial \chi_i}{\partial t} = \begin{cases} 0, & \theta < \theta_{\text{thr},i}, \\ A_i \exp\left(-\frac{E_i}{R\theta}\right) \theta^{n_i} (1 - \chi_i)^{m_i}, & \theta \geq \theta_{\text{thr},i}, \end{cases} \quad i = 1, \dots, N_r, \quad (1)$$

subject to

$$0 \leq \chi_i \leq 1. \quad (2)$$

The total pyrolysis source is the sum of the per-reaction contributions,

$$\pi_{\text{tot}} = \sum_{i=1}^{N_r} \pi_i, \quad (3)$$

with

$$\pi_i = \rho_{I,\sigma(i)} \varepsilon_{I,\sigma(i)} F_i \frac{\partial \chi_i}{\partial t}, \quad i = 1, \dots, N_r, \quad (4)$$

where χ_i is the reaction extent for reaction i , F_i is its mass-loss fraction, and $\sigma(i)$ is the solid phase to which reaction i is attached. The variable χ_i is a dimensionless progress variable, where $\chi_i = 0$ denotes the unreacted state for reaction i and $\chi_i = 1$ denotes a fully completed reaction.

3 Conservation of Solid Mass

$$\frac{\partial \rho_s}{\partial t} = -\pi_{\text{tot}} \quad (5)$$

4 Conservation of Gas Mass

$$\frac{\partial}{\partial t} (\phi \rho_g) + \nabla \cdot (\phi \rho_g \mathbf{v}_g) - \pi_{\text{tot}} = 0 \quad (6)$$

5 Conservation of Energy

$$\frac{\partial}{\partial t} (\rho e) + \nabla \cdot (\mathbf{q} + \mathbf{q}_{\text{conv}}) = 0 \quad (7)$$

6 Material Model

Ideal gas law for the pyrolysis gas:

$$\rho_g = \frac{p M(\theta, p)}{R \theta} \quad (8)$$

Darcy flow for the gas velocity:

$$\mathbf{v}_g = -\frac{K}{\mu_g \phi} \nabla p \quad (9)$$

Fourier's law for the conductive heat flux:

$$\mathbf{q} = -\kappa \nabla \theta \quad (10)$$

Convective heat flux for the gas phase:

$$\mathbf{q}_{\text{conv}} = h_g \phi \rho_g \mathbf{v}_g \quad (11)$$

Here $M(\theta, p)$, $h_g(\theta, p)$, and $\mu_g(\theta, p)$ denote the pyrolysis-gas molecular weight, specific enthalpy, and dynamic viscosity.

Internal energy of the mixture:

$$\rho e = \rho_s e_s + \phi \rho_g e_g \quad (12)$$

Degree of Char:

$$\beta = \frac{\rho_s - \rho_v}{\rho_c - \rho_v} \quad (13)$$

Here ρ_v and ρ_c denote the virgin and char solid densities, while $h_v(\theta, p)$, $h_c(\theta, p)$, $c_{pv}(\theta, p)$, $c_{pc}(\theta, p)$, $\kappa_v(\theta, p)$, and $\kappa_c(\theta, p)$ denote the corresponding virgin and char specific enthalpies, specific heat capacities, and thermal conductivities. Likewise, ϕ_v , ϕ_c , K_v , and K_c denote the virgin and char gas porosities and permeabilities.

Internal energy of the solid phase:

$$\rho_s e_s = (1 - \beta) \rho_v h_v + \beta \rho_c h_c \quad (14)$$

$$e_s = h_s = y_c h_c + y_v h_v \quad (15)$$

$$y_c = \frac{\beta \rho_c}{\rho_s}, \quad y_v = \frac{(1 - \beta) \rho_v}{\rho_s} \quad (16)$$

Constitutive closures for transport coefficients:

$$\phi(\beta) = (1 - \beta) \phi_v + \beta \phi_c \quad (17)$$

$$K(\beta) = (1 - \beta) K_v + \beta K_c \quad (18)$$

$$\kappa(\beta, \theta, p) = y_v \kappa_v(\theta, p) + y_c \kappa_c(\theta, p) \quad (19)$$

Internal energy of the gas phase:

$$e_g = h_g - p / \rho_g \quad (20)$$

Specific heat capacity of the solid phase:

$$\rho_s c_{ps} = \beta \rho_c c_{pc} + (1 - \beta) \rho_v c_{pv} \quad (21)$$

$$\frac{\partial}{\partial t} (\rho_s e_s) = \frac{\partial}{\partial t} (\rho_s h_s) = (\rho_c h_c - \rho_v h_v) \frac{\partial \beta}{\partial t} + ((1 - \beta) \rho_v c_{pv} + \beta \rho_c c_{pc}) \frac{\partial \theta}{\partial t} \quad (22)$$

$$\frac{\partial \beta}{\partial t} = \frac{1}{\rho_c - \rho_v} \frac{\partial \rho_s}{\partial t} \quad (23)$$

$$\frac{\partial}{\partial t} (\rho_s h_s) = \frac{(\rho_c h_c - \rho_v h_v)}{\rho_c - \rho_v} \frac{\partial \rho_s}{\partial t} + \rho_s c_{ps} \frac{\partial \theta}{\partial t} \quad (24)$$

$$\bar{h} = \frac{\rho_c h_c - \rho_v h_v}{\rho_c - \rho_v} \quad (25)$$

$$\frac{\partial}{\partial t} (\rho_s h_s) = \bar{h} \frac{\partial \rho_s}{\partial t} + \rho_s c_{ps} \frac{\partial \theta}{\partial t} = -\bar{h} \pi_{\text{tot}} + \rho_s c_{ps} \frac{\partial \theta}{\partial t} \quad (26)$$

7 Boundary Conditions

For the time-dependent Dirichlet variant documented here, both primary unknowns are prescribed on the top boundary:

$$\theta = \theta_D(t), \quad p = p_D(t) \quad \text{on } \Gamma_{\text{top}}. \quad (27)$$

In the test-driver interpretation, $\theta_D(t)$ is supplied by an external temperature schedule and $p_D(t)$ is supplied by the pressure schedule. No surface energy-balance flux law is imposed on Γ_{top} in this variant, because the top boundary values are enforced directly as essential time-dependent Dirichlet data.

Any surface recession in this variant is likewise prescribed kinematically through the ALE boundary motion, rather than being obtained from a separate surface recession or mass-balance closure.

On the remaining boundaries, writing the gas mass flux as

$$\mathbf{m}_g = \phi \rho_g \mathbf{v}_g, \quad (28)$$

the equations retain their natural conditions:

$$\mathbf{n} \cdot \mathbf{m}_g = 0, \quad \mathbf{n} \cdot (-\kappa \nabla \theta + h_g \mathbf{m}_g) = 0 \quad \text{on } \partial\Omega \setminus \Gamma_{\text{top}}. \quad (29)$$

8 Dirichlet Data

The time dependence is carried entirely by the prescribed boundary data

$$\theta_D = \theta_D(t), \quad p_D = p_D(t). \quad (30)$$

In particular, the strong form in the bulk is unchanged, and the boundary input enters only through the essential constraints applied at the top boundary.

9 ALE Definitions

For any Eulerian field $a(\mathbf{x}, t)$, its pullback to the reference domain is denoted by $\widehat{a}(\widehat{\mathbf{x}}, t) = a(\mathcal{A}(\widehat{\mathbf{x}}, t), t)$. When a spatial differential operator appears inside a hat, such as $\widehat{\nabla}\theta$ or $\widehat{\nabla} \cdot \mathbf{w}$, the meaning is: first evaluate the derivative in the current configuration, then pull the resulting field back to the reference domain. These objects are therefore not identified with reference-domain derivatives until the explicit ALE transformation formulas below are applied.

$$\mathcal{A} : \widehat{\Omega} \times [0, t_f] \rightarrow \Omega(t), \quad \mathbf{x} = \mathcal{A}(\widehat{\mathbf{x}}, t) \quad (31)$$

$$\mathbf{w}(\mathbf{x}, t) = \left. \frac{\partial \mathcal{A}}{\partial t} \right|_{\widehat{\mathbf{x}}}, \quad \widehat{\mathbf{w}}(\widehat{\mathbf{x}}, t) = \mathbf{w}(\mathcal{A}(\widehat{\mathbf{x}}, t), t) \quad (32)$$

$$\widehat{\mathbf{G}} = \frac{\partial \mathbf{x}}{\partial \widehat{\mathbf{x}}}, \quad \widehat{J} = \det(\widehat{\mathbf{G}}), \quad \widehat{\mathbf{F}} = \text{cof}(\widehat{\mathbf{G}}) = \widehat{J} \widehat{\mathbf{G}}^{-T} \quad (33)$$

$$\left. \frac{\partial \psi}{\partial t} \right|_{\widehat{\mathbf{x}}} = \frac{\partial \psi}{\partial t} + \mathbf{w} \cdot \nabla \psi \quad (34)$$

$$\widehat{\nabla} \psi = \frac{1}{\widehat{J}} \widehat{\mathbf{F}}^T \widehat{\nabla} \psi \quad (35)$$

$$\widehat{\nabla} \cdot \psi = \frac{1}{\widehat{J}} \widehat{\nabla} \cdot (\widehat{\mathbf{F}} \psi) \quad (36)$$

$$\frac{\partial \widehat{J}}{\partial t} = \widehat{J} \widehat{\nabla} \cdot \mathbf{w} \quad (37)$$

$$\left. \frac{\partial}{\partial t} (\widehat{\theta} \widehat{J}) \right|_{\widehat{\mathbf{x}}} = \left. \frac{\partial \widehat{\theta}}{\partial t} \right|_{\widehat{\mathbf{x}}} \widehat{J} + \widehat{\theta} \left. \frac{\partial \widehat{J}}{\partial t} \right|_{\widehat{\mathbf{x}}} = \left. \frac{\partial \widehat{\theta}}{\partial t} \right|_{\widehat{\mathbf{x}}} \widehat{J} + \widehat{\theta} \widehat{\nabla} \cdot \mathbf{w} \quad (38)$$

10 Final Weak Forms

$$\begin{aligned} \frac{d}{dt} \int_{\widehat{\Omega}} \widehat{\psi} \widehat{\phi} \widehat{\rho}_g \widehat{J} d\widehat{V} - \int_{\widehat{\Omega}} \widehat{\psi} \widehat{\phi} \widehat{\rho}_g \widehat{\nabla} \cdot (\widehat{\mathbf{F}} \widehat{\mathbf{w}}) d\widehat{V} - \int_{\widehat{\Omega}} \widehat{\psi} \widehat{\mathbf{F}} \widehat{\mathbf{w}} \cdot \widehat{\nabla} (\widehat{\phi} \widehat{\rho}_g) d\widehat{V} \\ + \int_{\widehat{\Omega}} \left(\widehat{\rho}_g \frac{\widehat{K}}{\widehat{\mu}_g} \frac{1}{\widehat{J}} \widehat{\mathbf{F}} \widehat{\mathbf{F}}^T \right) \widehat{\nabla} \widehat{p} \cdot \widehat{\nabla} \widehat{\psi} d\widehat{V} - \int_{\widehat{\Omega}} \widehat{\psi} \widehat{\pi}_{\text{tot}} \widehat{J} d\widehat{V} = 0 \end{aligned} \quad (39)$$

$$\begin{aligned} \int_{\widehat{\Omega}} \widehat{\psi} \widehat{\rho}_s \widehat{c}_{ps} \left. \frac{\partial}{\partial t} (\widehat{\theta} \widehat{J}) \right|_{\widehat{\mathbf{x}}} d\widehat{V} + \frac{d}{dt} \int_{\widehat{\Omega}} \widehat{\psi} \widehat{\phi} (\widehat{\rho}_g \widehat{h}_g - \widehat{p}) \widehat{J} d\widehat{V} \\ - \int_{\widehat{\Omega}} \widehat{\psi} \widehat{\rho}_s \widehat{c}_{ps} \widehat{\nabla} \cdot (\widehat{\mathbf{F}} \widehat{\theta} \widehat{\mathbf{w}}) d\widehat{V} - \int_{\widehat{\Omega}} \widehat{\psi} (\widehat{\phi} \widehat{\rho}_g \widehat{h}_g - \widehat{\phi} \widehat{p}) \widehat{\nabla} \cdot (\widehat{\mathbf{F}} \widehat{\mathbf{w}}) d\widehat{V} - \int_{\widehat{\Omega}} \widehat{\psi} \widehat{\mathbf{F}} \widehat{\mathbf{w}} \cdot \widehat{\nabla} (\widehat{\phi} (\widehat{\rho}_g \widehat{h}_g - \widehat{p})) d\widehat{V} \\ - \int_{\widehat{\Omega}} \widehat{\psi} \widehat{h} \widehat{\pi}_{\text{tot}} \widehat{J} d\widehat{V} + \int_{\widehat{\Omega}} \frac{1}{\widehat{J}} \widehat{\kappa} \widehat{\mathbf{F}} \widehat{\mathbf{F}}^T \widehat{\nabla} \widehat{\theta} \cdot \widehat{\nabla} \widehat{\psi} d\widehat{V} + \int_{\widehat{\Omega}} \frac{1}{\widehat{J}} \left(\widehat{h}_g \widehat{\rho}_g \frac{\widehat{K}}{\widehat{\mu}_g} \right) \widehat{\mathbf{F}} \widehat{\mathbf{F}}^T \widehat{\nabla} \widehat{p} \cdot \widehat{\nabla} \widehat{\psi} d\widehat{V} = 0. \end{aligned} \quad (40)$$

11 Derivation of ALE Mass Weak form

$$\frac{\partial}{\partial t} (\phi \rho_g) + \nabla \cdot (\phi \rho_g \mathbf{v}_g) - \pi_{\text{tot}} = 0 \quad (41)$$

$$\left. \frac{\partial}{\partial t} (\phi \rho_g) \right|_{\hat{\mathbf{x}}} - \mathbf{w} \cdot \nabla (\phi \rho_g) + \nabla \cdot (\phi \rho_g \mathbf{v}_g) - \pi_{\text{tot}} = 0 \quad (42)$$

$$\int_{\Omega} \psi \left. \frac{\partial}{\partial t} (\phi \rho_g) \right|_{\hat{\mathbf{x}}} dV - \int_{\Omega} \psi \mathbf{w} \cdot \nabla (\phi \rho_g) dV + \int_{\Omega} \psi \nabla \cdot (\phi \rho_g \mathbf{v}_g) dV - \int_{\Omega} \psi \pi_{\text{tot}} dV = 0 \quad (43)$$

$$\int_{\Omega} \left. \frac{\partial}{\partial t} (\psi \phi \rho_g) \right|_{\hat{\mathbf{x}}} dV - \int_{\Omega} \psi \mathbf{w} \cdot \nabla (\phi \rho_g) dV + \int_{\Omega} \psi \nabla \cdot (\phi \rho_g \mathbf{v}_g) dV - \int_{\Omega} \psi \pi_{\text{tot}} dV = 0 \quad (44)$$

$$\frac{d}{dt} \int_{\Omega} \psi \phi \rho_g dV - \int_{\Omega} \psi \phi \rho_g \nabla \cdot \mathbf{w} dV - \int_{\Omega} \psi \mathbf{w} \cdot \nabla (\phi \rho_g) dV + \int_{\Omega} \psi \nabla \cdot (\phi \rho_g \mathbf{v}_g) dV - \int_{\Omega} \psi \pi_{\text{tot}} dV = 0 \quad (45)$$

$$\frac{d}{dt} \int_{\Omega} \psi \phi \rho_g dV - \int_{\Omega} \psi \phi \rho_g \nabla \cdot \mathbf{w} dV - \int_{\Omega} \psi \mathbf{w} \cdot \nabla (\phi \rho_g) dV - \int_{\Omega} (\phi \rho_g \mathbf{v}_g) \cdot \nabla \psi dV - \int_{\Omega} \psi \pi_{\text{tot}} dV = 0 \quad (46)$$

$$\begin{aligned} \frac{d}{dt} \int_{\hat{\Omega}} \hat{\psi} \hat{\phi} \hat{\rho}_g \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\phi} \hat{\rho}_g \widehat{\nabla \cdot \mathbf{w}} \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\mathbf{w}} \cdot \widehat{\nabla (\phi \rho_g)} \hat{J} d\hat{V} \\ - \int_{\hat{\Omega}} \hat{\phi} \hat{\rho}_g \hat{\mathbf{v}}_g \cdot \widehat{\nabla \psi} \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\pi}_{\text{tot}} \hat{J} d\hat{V} = 0 \end{aligned} \quad (47)$$

$$\begin{aligned} \frac{d}{dt} \int_{\hat{\Omega}} \hat{\psi} \hat{\phi} \hat{\rho}_g \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\phi} \hat{\rho}_g \frac{1}{\hat{J}} \widehat{\nabla} \cdot (\hat{\mathbf{F}} \hat{\mathbf{w}}) \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\mathbf{w}} \cdot \left(\frac{1}{\hat{J}} \hat{\mathbf{F}}^T \widehat{\nabla} (\hat{\phi} \hat{\rho}_g) \right) \hat{J} d\hat{V} \\ - \int_{\hat{\Omega}} \hat{\phi} \hat{\rho}_g \hat{\mathbf{v}}_g \cdot \left(\frac{1}{\hat{J}} \hat{\mathbf{F}}^T \widehat{\nabla} \hat{\psi} \right) \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\pi}_{\text{tot}} \hat{J} d\hat{V} = 0 \end{aligned} \quad (48)$$

$$\hat{\mathbf{v}}_g = -\frac{\hat{K}}{\hat{\mu}_g \hat{\phi}} \widehat{\nabla} p = -\frac{\hat{K}}{\hat{\mu}_g \hat{\phi}} \frac{1}{\hat{J}} \hat{\mathbf{F}}^T \widehat{\nabla} \hat{p} \quad (49)$$

$$\begin{aligned} \frac{d}{dt} \int_{\hat{\Omega}} \hat{\psi} \hat{\phi} \hat{\rho}_g \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\phi} \hat{\rho}_g \frac{1}{\hat{J}} \widehat{\nabla} \cdot (\hat{\mathbf{F}} \hat{\mathbf{w}}) \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\mathbf{w}} \cdot \left(\frac{1}{\hat{J}} \hat{\mathbf{F}}^T \widehat{\nabla} (\hat{\phi} \hat{\rho}_g) \right) \hat{J} d\hat{V} \\ + \int_{\hat{\Omega}} \hat{\phi} \hat{\rho}_g \left(\frac{\hat{K}}{\hat{\mu}_g \hat{\phi}} \frac{1}{\hat{J}} \hat{\mathbf{F}}^T \widehat{\nabla} \hat{p} \right) \cdot \left(\frac{1}{\hat{J}} \hat{\mathbf{F}}^T \widehat{\nabla} \hat{\psi} \right) \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\pi}_{\text{tot}} \hat{J} d\hat{V} = 0 \end{aligned} \quad (50)$$

$$\begin{aligned} \frac{d}{dt} \int_{\hat{\Omega}} \hat{\psi} \hat{\phi} \hat{\rho}_g \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\phi} \hat{\rho}_g \widehat{\nabla} \cdot (\hat{\mathbf{F}} \hat{\mathbf{w}}) d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\mathbf{w}} \cdot (\hat{\mathbf{F}}^T \widehat{\nabla} (\hat{\phi} \hat{\rho}_g)) d\hat{V} \\ + \int_{\hat{\Omega}} \hat{\phi} \hat{\rho}_g \left(\frac{\hat{K}}{\hat{\mu}_g \hat{\phi}} \frac{1}{\hat{J}} \hat{\mathbf{F}}^T \widehat{\nabla} \hat{p} \right) \cdot (\hat{\mathbf{F}}^T \widehat{\nabla} \hat{\psi}) d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\pi}_{\text{tot}} \hat{J} d\hat{V} = 0 \end{aligned} \quad (51)$$

$$\begin{aligned} \frac{d}{dt} \int_{\hat{\Omega}} \hat{\psi} \hat{\phi} \hat{\rho}_g \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\phi} \hat{\rho}_g \hat{\nabla} \cdot (\hat{\mathbf{F}} \hat{\mathbf{w}}) d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\mathbf{F}} \hat{\mathbf{w}} \cdot \hat{\nabla} (\hat{\phi} \hat{\rho}_g) d\hat{V} \\ + \int_{\hat{\Omega}} \left(\hat{\rho}_g \frac{\hat{K}}{\hat{\mu}_g} \frac{1}{\hat{J}} \hat{\mathbf{F}} \hat{\mathbf{F}}^T \right) \hat{\nabla} \hat{p} \cdot \hat{\nabla} \hat{\psi} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\pi}_{\text{tot}} \hat{J} d\hat{V} = 0 \end{aligned} \quad (52)$$

12 Derivation of ALE Energy Weak form

$$\frac{\partial}{\partial t} (\rho e) + \nabla \cdot (-\kappa \nabla \theta + h_g \phi \rho_g \mathbf{v}_g) = 0. \quad (53)$$

$$\rho_s c_{ps} \frac{\partial \theta}{\partial t} + \frac{\partial}{\partial t} (\phi(\rho_g h_g - p)) - \bar{h} \pi_{\text{tot}} + \nabla \cdot (-\kappa \nabla \theta + h_g \phi \rho_g \mathbf{v}_g) = 0. \quad (54)$$

$$\rho_s c_{ps} \left(\frac{\partial \theta}{\partial t} \Big|_{\hat{\mathbf{x}}} - \mathbf{w} \cdot \nabla \theta \right) + \frac{\partial}{\partial t} (\phi(\rho_g h_g - p)) \Big|_{\hat{\mathbf{x}}} - \mathbf{w} \cdot \nabla (\phi(\rho_g h_g - p)) - \bar{h} \pi_{\text{tot}} + \nabla \cdot (-\kappa \nabla \theta + h_g \phi \rho_g \mathbf{v}_g) = 0. \quad (55)$$

$$\begin{aligned} \int_{\Omega} \psi \rho_s c_{ps} \left(\frac{\partial \theta}{\partial t} \Big|_{\hat{\mathbf{x}}} - \mathbf{w} \cdot \nabla \theta \right) dV + \int_{\Omega} \psi \frac{\partial}{\partial t} (\phi(\rho_g h_g - p)) \Big|_{\hat{\mathbf{x}}} dV \\ - \int_{\Omega} \psi \mathbf{w} \cdot \nabla (\phi(\rho_g h_g - p)) dV - \int_{\Omega} \psi \bar{h} \pi_{\text{tot}} dV + \int_{\Omega} \psi \nabla \cdot (-\kappa \nabla \theta + h_g \phi \rho_g \mathbf{v}_g) dV = 0. \end{aligned} \quad (56)$$

$$\begin{aligned} \int_{\Omega} \psi \rho_s c_{ps} \left(\frac{\partial \theta}{\partial t} \Big|_{\hat{\mathbf{x}}} - \mathbf{w} \cdot \nabla \theta \right) dV + \int_{\Omega} \frac{\partial}{\partial t} (\psi \phi(\rho_g h_g - p)) \Big|_{\hat{\mathbf{x}}} dV \\ - \int_{\Omega} \psi \mathbf{w} \cdot \nabla (\phi(\rho_g h_g - p)) dV - \int_{\Omega} \psi \bar{h} \pi_{\text{tot}} dV + \int_{\Omega} \psi \nabla \cdot (-\kappa \nabla \theta + h_g \phi \rho_g \mathbf{v}_g) dV = 0. \end{aligned} \quad (57)$$

$$\begin{aligned} \int_{\Omega} \psi \rho_s c_{ps} \left(\frac{1}{\hat{J}} \frac{\partial}{\partial t} (\theta \hat{J}) \Big|_{\hat{\mathbf{x}}} - \theta \nabla \cdot \mathbf{w} - \mathbf{w} \cdot \nabla \theta \right) dV + \int_{\Omega} \frac{\partial}{\partial t} (\psi \phi(\rho_g h_g - p)) \Big|_{\hat{\mathbf{x}}} dV \\ - \int_{\Omega} \psi \mathbf{w} \cdot \nabla (\phi(\rho_g h_g - p)) dV - \int_{\Omega} \psi \bar{h} \pi_{\text{tot}} dV + \int_{\Omega} \psi \nabla \cdot (-\kappa \nabla \theta + h_g \phi \rho_g \mathbf{v}_g) dV = 0. \end{aligned} \quad (58)$$

$$\begin{aligned} \int_{\Omega} \psi \rho_s c_{ps} \left(\frac{1}{\hat{J}} \frac{\partial}{\partial t} (\theta \hat{J}) \Big|_{\hat{\mathbf{x}}} - \nabla \cdot (\theta \mathbf{w}) \right) dV + \int_{\Omega} \frac{\partial}{\partial t} (\psi \phi(\rho_g h_g - p)) \Big|_{\hat{\mathbf{x}}} dV \\ - \int_{\Omega} \psi \mathbf{w} \cdot \nabla (\phi(\rho_g h_g - p)) dV - \int_{\Omega} \psi \bar{h} \pi_{\text{tot}} dV + \int_{\Omega} \psi \nabla \cdot (-\kappa \nabla \theta + h_g \phi \rho_g \mathbf{v}_g) dV = 0. \end{aligned} \quad (59)$$

$$\begin{aligned} \int_{\Omega} \psi \rho_s c_{ps} \left(\frac{1}{\hat{J}} \frac{\partial}{\partial t} (\theta \hat{J}) \Big|_{\hat{\mathbf{x}}} - \nabla \cdot (\theta \mathbf{w}) \right) dV + \frac{d}{dt} \int_{\Omega} \psi \phi(\rho_g h_g - p) dV - \int_{\Omega} \psi (\phi \rho_g h_g - \phi p) \nabla \cdot \mathbf{w} dV \\ - \int_{\Omega} \psi \mathbf{w} \cdot \nabla (\phi(\rho_g h_g - p)) dV - \int_{\Omega} \psi \bar{h} \pi_{\text{tot}} dV + \int_{\Omega} \psi \nabla \cdot (-\kappa \nabla \theta + h_g \phi \rho_g \mathbf{v}_g) dV = 0. \end{aligned} \quad (60)$$

$$\begin{aligned}
& \int_{\Omega} \psi \rho_s c_{ps} \frac{1}{\hat{J}} \frac{\partial}{\partial t} \left(\theta \hat{J} \right) \Big|_{\hat{\mathbf{x}}} dV + \frac{d}{dt} \int_{\Omega} \psi \phi (\rho_g h_g - p) dV \\
& - \int_{\Omega} \psi \rho_s c_{ps} \nabla \cdot (\theta \mathbf{w}) dV - \int_{\Omega} \psi (\phi \rho_g h_g - \phi p) \nabla \cdot \mathbf{w} dV - \int_{\Omega} \psi \mathbf{w} \cdot \nabla (\phi (\rho_g h_g - p)) dV \\
& - \int_{\Omega} \psi \bar{h} \pi_{\text{tot}} dV + \int_{\Omega} \psi \nabla \cdot (-\kappa \nabla \theta + h_g \phi \rho_g \mathbf{v}_g) dV = 0. \quad (61)
\end{aligned}$$

$$\begin{aligned}
& \int_{\Omega} \psi \rho_s c_{ps} \frac{1}{\hat{J}} \frac{\partial}{\partial t} \left(\theta \hat{J} \right) \Big|_{\hat{\mathbf{x}}} dV + \frac{d}{dt} \int_{\Omega} \psi \phi (\rho_g h_g - p) dV \\
& - \int_{\Omega} \psi \rho_s c_{ps} \nabla \cdot (\theta \mathbf{w}) dV - \int_{\Omega} \psi (\phi \rho_g h_g - \phi p) \nabla \cdot \mathbf{w} dV - \int_{\Omega} \psi \mathbf{w} \cdot \nabla (\phi (\rho_g h_g - p)) dV \\
& - \int_{\Omega} \psi \bar{h} \pi_{\text{tot}} dV - \int_{\Omega} (-\kappa \nabla \theta + h_g \phi \rho_g \mathbf{v}_g) \cdot \nabla \psi dV = 0. \quad (62)
\end{aligned}$$

$$\begin{aligned}
& \int_{\hat{\Omega}} \hat{\psi} \hat{\rho}_s \hat{c}_{ps} \frac{1}{\hat{J}} \frac{\partial}{\partial t} \left(\hat{\theta} \hat{J} \right) \Big|_{\hat{\mathbf{x}}} \hat{J} d\hat{V} + \frac{d}{dt} \int_{\hat{\Omega}} \hat{\psi} \hat{\phi} (\hat{\rho}_g \hat{h}_g - \hat{p}) \hat{J} d\hat{V} \\
& - \int_{\hat{\Omega}} \hat{\psi} \hat{\rho}_s \hat{c}_{ps} \widehat{\nabla \cdot (\theta \mathbf{w})} \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \left(\hat{\phi} \hat{\rho}_g \hat{h}_g - \hat{\phi} \hat{p} \right) \widehat{\nabla \cdot \mathbf{w}} \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\mathbf{w}} \cdot \nabla (\phi (\hat{\rho}_g \hat{h}_g - \hat{p})) \hat{J} d\hat{V} \\
& - \int_{\hat{\Omega}} \hat{\psi} \hat{h} \hat{\pi}_{\text{tot}} \hat{J} d\hat{V} - \int_{\hat{\Omega}} \left(-\hat{\kappa} \widehat{\nabla \theta} + \hat{h}_g \hat{\phi} \hat{\rho}_g \hat{\mathbf{v}}_g \right) \cdot \widehat{\nabla \psi} \hat{J} d\hat{V} = 0. \quad (63)
\end{aligned}$$

$$\begin{aligned}
& \int_{\hat{\Omega}} \hat{\psi} \hat{\rho}_s \hat{c}_{ps} \frac{1}{\hat{J}} \frac{\partial}{\partial t} \left(\hat{\theta} \hat{J} \right) \Big|_{\hat{\mathbf{x}}} \hat{J} d\hat{V} + \frac{d}{dt} \int_{\hat{\Omega}} \hat{\psi} \hat{\phi} (\hat{\rho}_g \hat{h}_g - \hat{p}) \hat{J} d\hat{V} \\
& - \int_{\hat{\Omega}} \hat{\psi} \hat{\rho}_s \hat{c}_{ps} \frac{1}{\hat{J}} \widehat{\nabla \cdot (\hat{\mathbf{F}} \hat{\theta} \hat{\mathbf{w}})} \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \left(\hat{\phi} \hat{\rho}_g \hat{h}_g - \hat{\phi} \hat{p} \right) \frac{1}{\hat{J}} \widehat{\nabla \cdot (\hat{\mathbf{F}} \hat{\mathbf{w}})} \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\mathbf{w}} \cdot \left(\frac{1}{\hat{J}} \hat{\mathbf{F}}^T \widehat{\nabla} \left(\hat{\phi} (\hat{\rho}_g \hat{h}_g - \hat{p}) \right) \right) \hat{J} d\hat{V} \\
& - \int_{\hat{\Omega}} \hat{\psi} \hat{h} \hat{\pi}_{\text{tot}} \hat{J} d\hat{V} - \int_{\hat{\Omega}} \left(-\hat{\kappa} \left(\frac{1}{\hat{J}} \hat{\mathbf{F}}^T \widehat{\nabla} \hat{\theta} \right) + \hat{h}_g \hat{\phi} \hat{\rho}_g \left(-\frac{\hat{K}}{\hat{\mu}_g \hat{\phi}} \frac{1}{\hat{J}} \hat{\mathbf{F}}^T \widehat{\nabla} \hat{p} \right) \right) \cdot \left(\frac{1}{\hat{J}} \hat{\mathbf{F}}^T \widehat{\nabla} \hat{\psi} \right) \hat{J} d\hat{V} = 0. \quad (64)
\end{aligned}$$

$$\begin{aligned}
& \int_{\hat{\Omega}} \hat{\psi} \hat{\rho}_s \hat{c}_{ps} \frac{1}{\hat{J}} \frac{\partial}{\partial t} \left(\hat{\theta} \hat{J} \right) \Big|_{\hat{\mathbf{x}}} \hat{J} d\hat{V} + \frac{d}{dt} \int_{\hat{\Omega}} \hat{\psi} \hat{\phi} (\hat{\rho}_g \hat{h}_g - \hat{p}) \hat{J} d\hat{V} \\
& - \int_{\hat{\Omega}} \hat{\psi} \hat{\rho}_s \hat{c}_{ps} \frac{1}{\hat{J}} \widehat{\nabla \cdot (\hat{\mathbf{F}} \hat{\theta} \hat{\mathbf{w}})} \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \left(\hat{\phi} \hat{\rho}_g \hat{h}_g - \hat{\phi} \hat{p} \right) \frac{1}{\hat{J}} \widehat{\nabla \cdot (\hat{\mathbf{F}} \hat{\mathbf{w}})} \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\mathbf{w}} \cdot \left(\frac{1}{\hat{J}} \hat{\mathbf{F}}^T \widehat{\nabla} \left(\hat{\phi} (\hat{\rho}_g \hat{h}_g - \hat{p}) \right) \right) \hat{J} d\hat{V} \\
& - \int_{\hat{\Omega}} \hat{\psi} \hat{h} \hat{\pi}_{\text{tot}} \hat{J} d\hat{V} - \int_{\hat{\Omega}} \left(-\hat{\kappa} \left(\frac{1}{\hat{J}} \hat{\mathbf{F}}^T \widehat{\nabla} \hat{\theta} \right) + \hat{h}_g \hat{\phi} \hat{\rho}_g \left(-\frac{\hat{K}}{\hat{\mu}_g \hat{\phi}} \frac{1}{\hat{J}} \hat{\mathbf{F}}^T \widehat{\nabla} \hat{p} \right) \right) \cdot \left(\frac{1}{\hat{J}} \hat{\mathbf{F}}^T \widehat{\nabla} \hat{\psi} \right) \hat{J} d\hat{V} = 0. \quad (65)
\end{aligned}$$

$$\begin{aligned}
& \int_{\hat{\Omega}} \hat{\psi} \hat{\rho}_s \hat{c}_{ps} \frac{1}{\hat{J}} \frac{\partial}{\partial t} \left(\hat{\theta} \hat{J} \right) \Big|_{\hat{\mathbf{x}}} \hat{J} d\hat{V} + \frac{d}{dt} \int_{\hat{\Omega}} \hat{\psi} \hat{\phi} (\hat{\rho}_g \hat{h}_g - \hat{p}) \hat{J} d\hat{V} \\
& - \int_{\hat{\Omega}} \hat{\psi} \hat{\rho}_s \hat{c}_{ps} \frac{1}{\hat{J}} \hat{\nabla} \cdot (\hat{\mathbf{F}} \hat{\theta} \hat{\mathbf{w}}) \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} (\hat{\phi} \hat{\rho}_g \hat{h}_g - \hat{\phi} \hat{p}) \frac{1}{\hat{J}} \hat{\nabla} \cdot (\hat{\mathbf{F}} \hat{\mathbf{w}}) \hat{J} d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\mathbf{w}} \cdot \left(\frac{1}{\hat{J}} \hat{\mathbf{F}}^T \hat{\nabla} (\hat{\phi} (\hat{\rho}_g \hat{h}_g - \hat{p})) \right) \hat{J} d\hat{V} \\
& - \int_{\hat{\Omega}} \hat{\psi} \hat{h} \hat{\pi}_{\text{tot}} \hat{J} d\hat{V} - \int_{\hat{\Omega}} \left(-\hat{\kappa} \left(\frac{1}{\hat{J}} \hat{\mathbf{F}}^T \hat{\nabla} \hat{\theta} \right) \right) \cdot \left(\frac{1}{\hat{J}} \hat{\mathbf{F}}^T \hat{\nabla} \hat{\psi} \right) \hat{J} d\hat{V} - \int_{\hat{\Omega}} \left(\hat{h}_g \hat{\phi} \hat{\rho}_g \left(-\frac{\hat{K}}{\hat{\mu}_g \hat{\phi}} \frac{1}{\hat{J}} \hat{\mathbf{F}}^T \hat{\nabla} \hat{p} \right) \right) \cdot \left(\frac{1}{\hat{J}} \hat{\mathbf{F}}^T \hat{\nabla} \hat{\psi} \right) \hat{J} d\hat{V} = 0.
\end{aligned} \tag{66}$$

$$\begin{aligned}
& \int_{\hat{\Omega}} \hat{\psi} \hat{\rho}_s \hat{c}_{ps} \frac{\partial}{\partial t} \left(\hat{\theta} \hat{J} \right) \Big|_{\hat{\mathbf{x}}} d\hat{V} + \frac{d}{dt} \int_{\hat{\Omega}} \hat{\psi} \hat{\phi} (\hat{\rho}_g \hat{h}_g - \hat{p}) \hat{J} d\hat{V} \\
& - \int_{\hat{\Omega}} \hat{\psi} \hat{\rho}_s \hat{c}_{ps} \hat{\nabla} \cdot (\hat{\mathbf{F}} \hat{\theta} \hat{\mathbf{w}}) d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} (\hat{\phi} \hat{\rho}_g \hat{h}_g - \hat{\phi} \hat{p}) \hat{\nabla} \cdot (\hat{\mathbf{F}} \hat{\mathbf{w}}) d\hat{V} - \int_{\hat{\Omega}} \hat{\psi} \hat{\mathbf{F}} \hat{\mathbf{w}} \cdot \hat{\nabla} (\hat{\phi} (\hat{\rho}_g \hat{h}_g - \hat{p})) d\hat{V} \\
& - \int_{\hat{\Omega}} \hat{\psi} \hat{h} \hat{\pi}_{\text{tot}} \hat{J} d\hat{V} + \int_{\hat{\Omega}} \frac{1}{\hat{J}} \hat{\kappa} \hat{\mathbf{F}} \hat{\mathbf{F}}^T \hat{\nabla} \hat{\theta} \cdot \hat{\nabla} \hat{\psi} d\hat{V} + \int_{\hat{\Omega}} \frac{1}{\hat{J}} \left(\hat{h}_g \hat{\rho}_g \frac{\hat{K}}{\hat{\mu}_g} \right) \hat{\mathbf{F}} \hat{\mathbf{F}}^T \hat{\nabla} \hat{p} \cdot \hat{\nabla} \hat{\psi} d\hat{V} = 0. \tag{67}
\end{aligned}$$